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Completed
11/03/26

verification of kirchoff's voltage law and kirchoff's current law

AIM

To verify kirchoff's voltage law for loop 1 and kirchoff's current law in node A for the following circuit

Apparatus Required

laptop with protous software

SAVEETHA
ENGINEERING COLLEGE
SAVEETHA NAGAR 602 105
BE THE BEST
2001

THEORY

According to kirchoff's voltage law sum of potential raises are equal to sum of potential drop in a loop

$$V = V_1 + V_2 + V_3$$

$$100 = IR_1 + IR_2 + IR_3$$

$$R_{eq} = R1 + R2 + R3$$

$$R_{eq} = 30 + 50 + 100$$

$$R_{eq} = 180\,\Omega$$

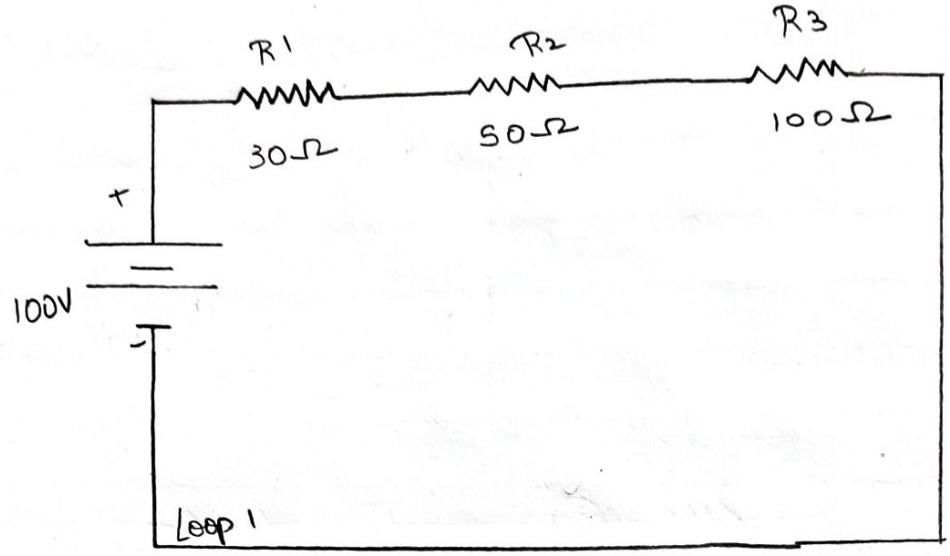
$$I = \frac{V}{R_{eq}} = \frac{100}{180} = 0.556\,A$$

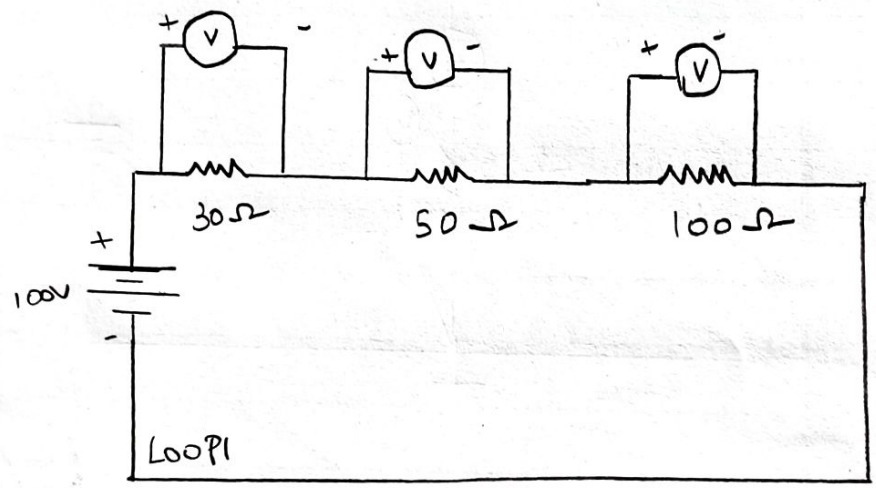
$V1 = V_{30\,\Omega} = IR_1 = 0.556 \times 30 = 16.68\,V$

$V_2 = V_{50\,\Omega} = IR_2 = 0.556 \times 50 = 27.8\,V$

$V_3 = V_{100\,\Omega} = IR_3 = 0.556 \times 100 = 55.6\,V$

Circuit





KVL	source	$V_{30\Omega}$	$V_{50\Omega}$	$V_{10\Omega}$	$V_{30\Omega} + V_{50\Omega} + V_{100\Omega}$ V
Theory	100V	16.68V	27.8V	55.6V	100.05
Practice	100V	16.7V	27.8V	55.6V	100.1

substitute V_1, V_2 and V_3 value in ①

$$100 = 16.68 + 27.8 + 55.6$$

$$100V = 100V$$

Hence proved

Kirchoff's current law

In a junction, incoming current is equal to outgoing current

At node A

$$I_{30\Omega} + I_{50\Omega} + I_{100\Omega} = 0$$

$$\frac{V-100}{30} + \frac{V}{50} + \frac{V}{100} = 0$$

$$V\left[\frac{1}{30} + \frac{1}{50} + \frac{1}{100}\right] - \frac{100}{30} = 0$$

$$V[0.0633] - 3.333 = 0$$

$$V = \frac{3.333}{0.0633}$$

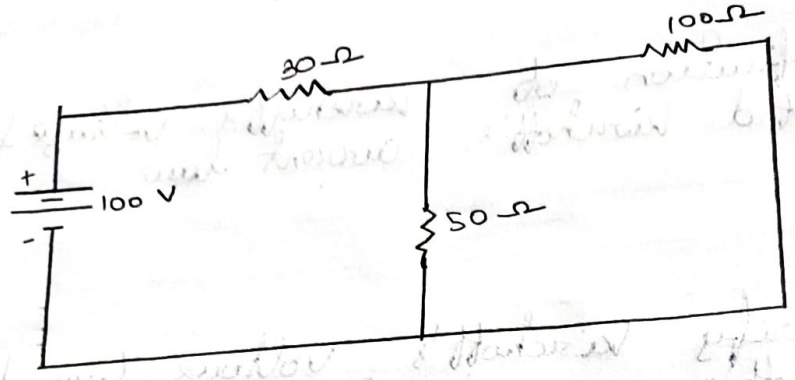
$$V = 52.63V$$

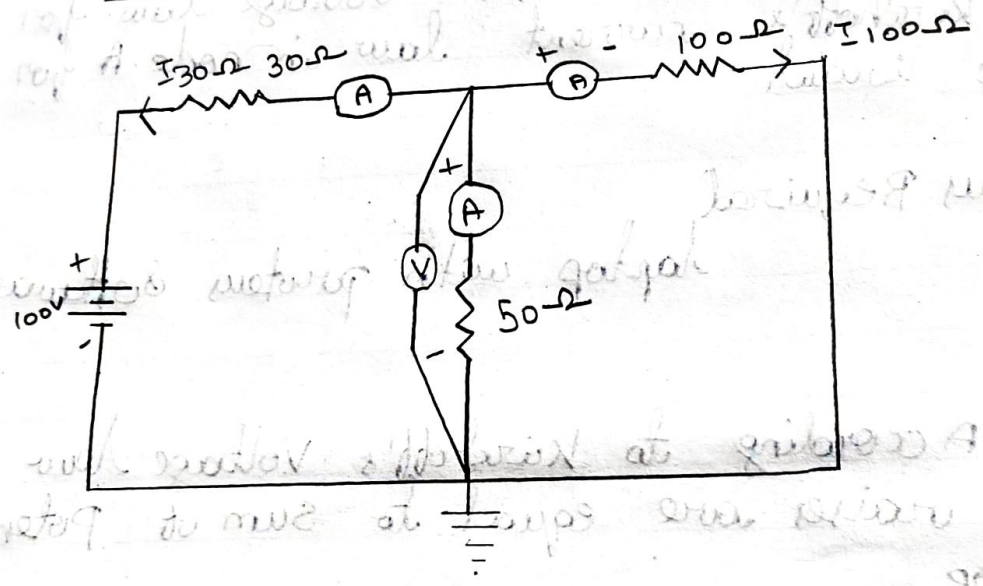
$$I_{30\Omega} = \frac{V-100}{30} = \frac{52.63-100}{30} = -1.38A$$

$$I_{50\Omega} = \frac{V}{50} = \frac{52.63}{50} = 1.05A$$

$$I_{100\Omega} = \frac{V}{100} = \frac{52.63}{100} = 0.53A$$

Circuit





Tabulation

KCL	I 30Ω	I 50Ω	I 100Ω	VA
Theoritical	1.48A	1.85A	0.56A	52.63V
Practical	1.58A	1.05A	0.53A	52.6V

At node A:

$$I_{30\Omega} + I_{50\Omega} + I_{100\Omega} = 0$$

$$-1.58 + 1.05 + 0.58 = 0$$

Hence Proved

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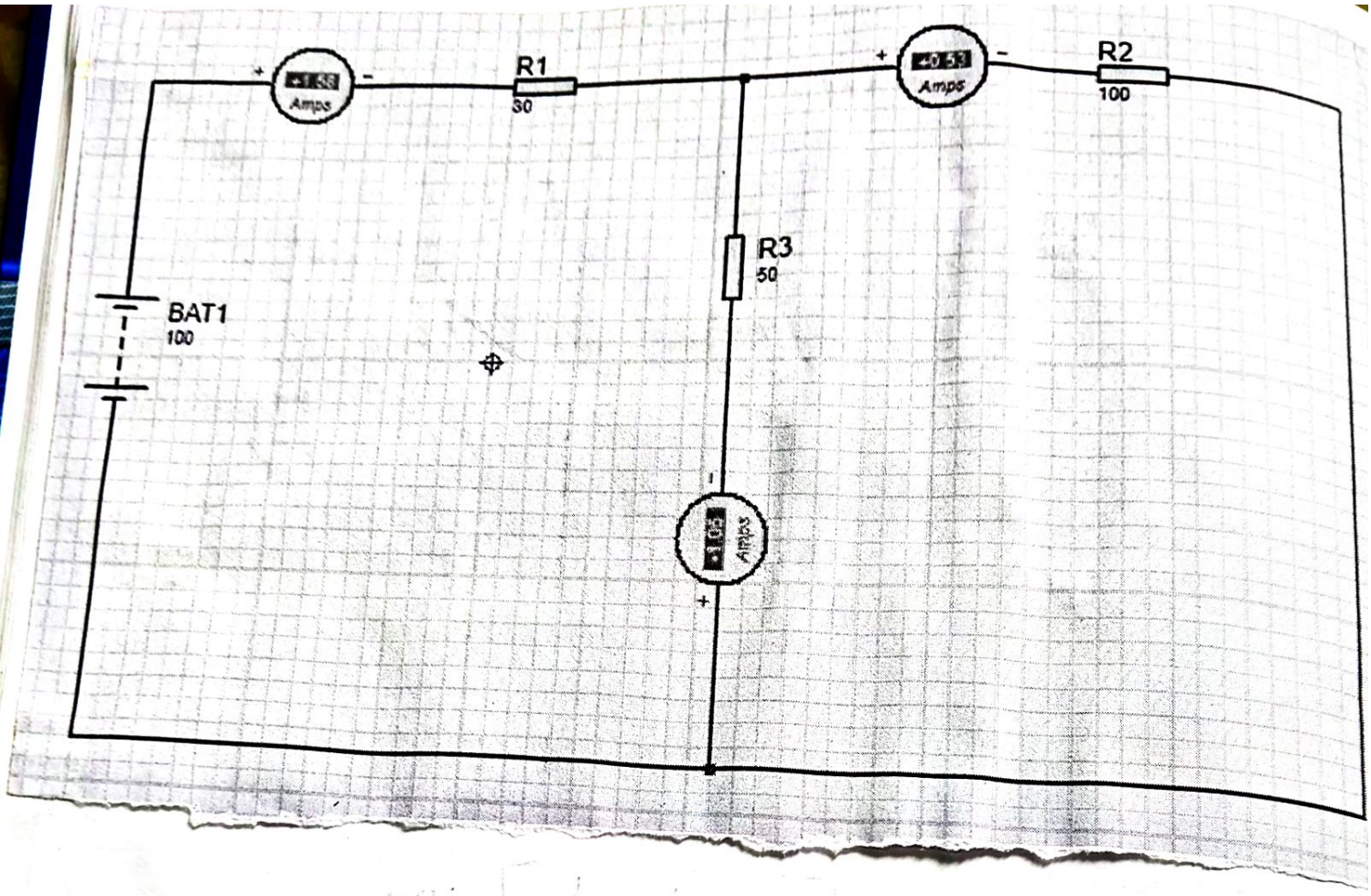
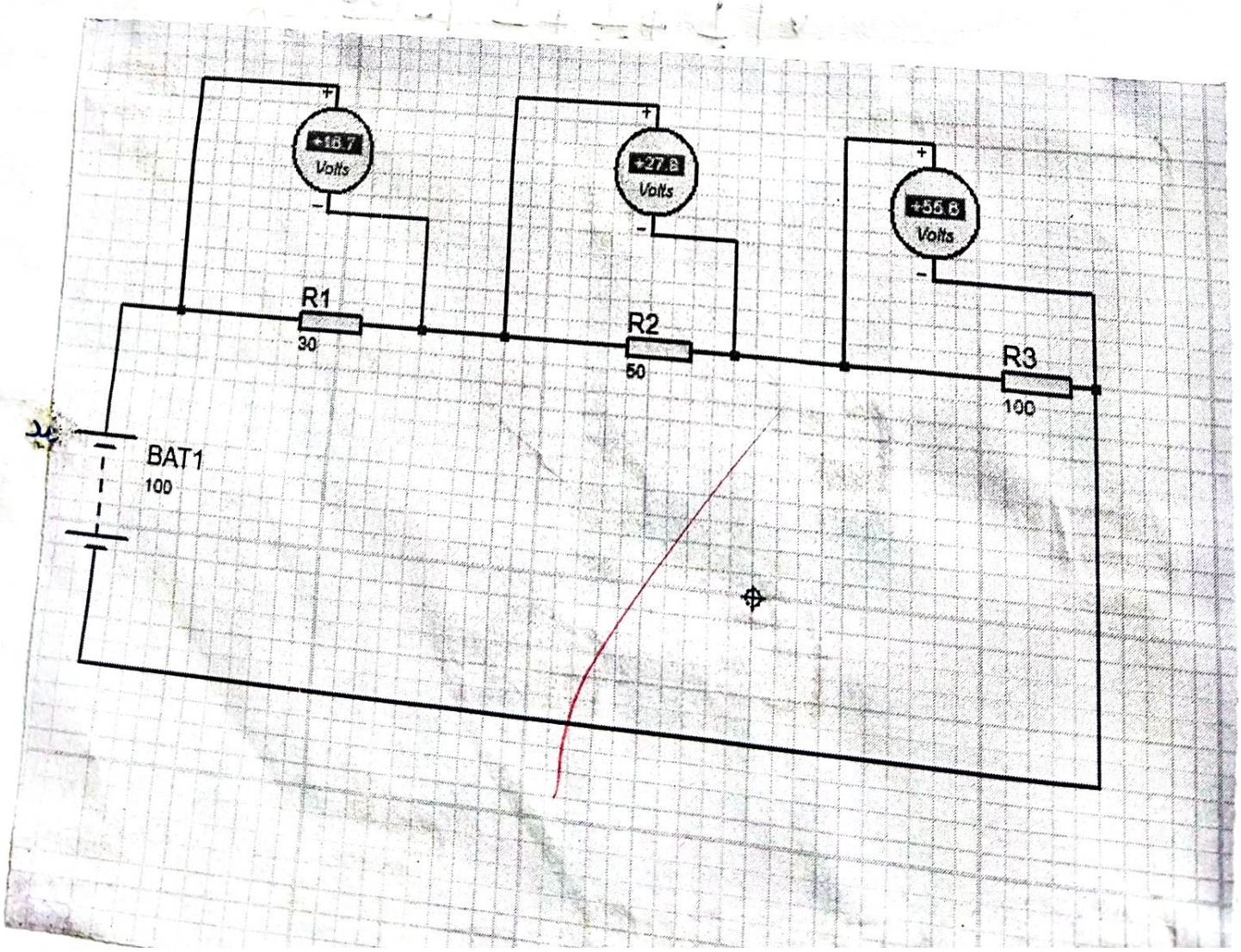
04/02/2026

Result

Thus, the kirchoff's voltage law and kirchoff's current law has been verified successfully


+1.33
Amps
R1
30
+0.53
Amps
R2
100
R3
50
BAT1
100
-1.05
Amps


+16.7
Volts
+27.8
Volts
+55.6
Volts
R1
30
R2
50
R3
100
BAT1
100